Additional Management Practice Assessment
Supporting Document IndigoCarbon_US-1_2023_0018

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Contents

1 Background **1**

2 Construction of the additional management practice assessment **1**

3 Subsequent years of the additional management practice assessment **4**

4 Additional management practice assessment spatial granularity **4**

5 Summary of Changes **5**

1 Background

Assessing additional management practices is a critical step to determine if a grower qualifies for the Carbon by Indigo program. In this appendix, the steps followed to assess if a grower adopted an additional management practice between the current management period (project scenario) and the historical management period (historical period) are detailed. The 'project-specific measures' used to quantify additional practices are described in IndigoCarbon_US-1_2023_0024a. The steps here present how grower input and cultivation cycles are used to assess for additionality, applying a similar approach used to construct and evaluate baseline threads. Collectively, these steps are implemented in the Carbon Data Pipeline, with Gate 2, Level 5 of the Grower Data QA/QC (described fully in IndigoCarbon_US-1_2023_0032) specifically assessing for additionality.

In this process, the growers in the Project attest to an additional management practice and the year it was adopted on each field. All potential project years are scanned for confirmed additional practice changes. This is done by comparing the data in the index cultivation cycle to the baseline threads defined in the historical period, and in years following the subsequent cultivation cycles are compared to the same baseline threads to determine if a practice change had occurred.

Please see the appendix materials for Cultivation cycle and crop growing season definition and Baseline thread construction (IndigoCarbon_US-1_2023_0016 and IndigoCarbon_US-1_2023_0048, respectively) for background materials on these terms.

2 Construction of the additional management practice assessment

Step 1 - Identify the proposed Additional management practice

In each year from 2018 forward, a grower selects the practices they changed or intend to change on each field through the Carbon by Indigo UI. The options presented to the grower for selection at this step are broad practice categories that are assessed with more detail in subsequent steps.

This practice selection helps define the minimum data requirements for each field. Indigo requires a completed field (including all minimum data requirements, field boundaries, field unique identification, etc.) to evaluate the adoption

of the stated additional management practices. The data is absorbed from the Carbon by Indigo UI and the index cultivation cycle is then determined.

Step 2 - Determine the index cultivation cycle

During Step 1, the grower attests to the cultivation cycle in which a practice change occurred. Indigo will assess this attestation and a field's additional practice changes during any potential cultivation cycles in the Project. If a grower implemented a practice change at any point during a potential cultivation cycle, then the earliest cultivation cycle was considered to be the index cultivation cycle.

Each field's historical period, which is used to define baseline threads, ends immediately before the beginning of the index cultivation cycle.

The earliest possible index cultivation cycle is the cultivation cycle that begins in 2018 but does not end prior to September 30, 2018 under guidance from the Soil Enrichment Protocol (SEP). If no practice change is detected for the 2018 cultivation cycle, then the code will iteratively assess practice changes beginning the day after the last cultivation cycle. See chart below (Figure 1) for examples of possible index cultivation cycles for the 2019 and 2020 vintage years of credit generation.

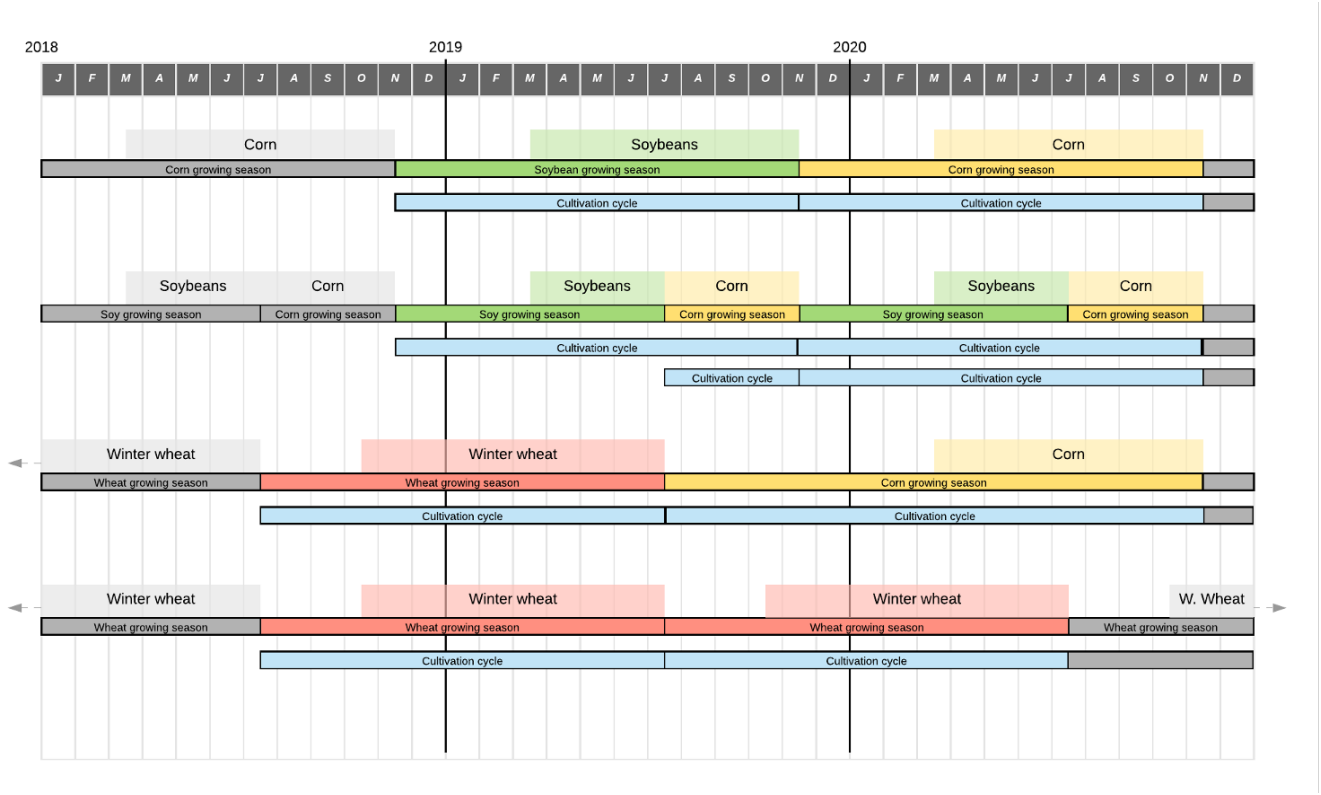


Figure 1: Examples of index cultivation

Note: Examples with multiple cultivation cycles reflect unique fields with different agronomic circumstances.

Step 3 - Quantify the stated additional management practice in the index cultivation cycle and the historical period

When determining how to quantify additional practices, the objective was to adopt an approach that was easy to communicate to growers and was aligned with other components of the methodology. As mentioned in Step 1, the additional management practices were defined based on the data collected through the Carbon by Indigo UI. Each definition of an additional management practice was assigned a quantification attribute to determine whether the change to the new management practice would be additional (which is evaluated in Step 4). Table 2 displays an example for how a practice would be quantified, which would take place in both the index cultivation cycle and the

appropriate historical comparison period. For a full summary of this logic, please request the published code for assessing additionality.

Table 1: Example of quantifying an additional management practice

Practice Category	**Additional Practice**	**Quantification**
Tillage and residue management	Tillage reduction through delayed tilling at minimum 90 days	Number of days between harvest and subsequent tillage event
Nitrogen Management	Addition of an enhanced efficiency nitrogen stabilizer	Change in nitrogen product add-ons
Crop planting and harvesting	Longer duration of cover crops through earlier planting or delayed termination at minimum 7 days	• Length of cover crop season • Date of cover crop termination

To quantify the stated additional management practice, the historical period must be determined. The approach utilized for this closely aligns with the approach used to construct and evaluate baseline threads: it depends on the crop grown during the index cultivation cycle and the attested additional management practice. In most cases, the additional management practice will be considered "crop-specific" and evaluated against a cash-crop-specific historical period. The process is described with additional detail in table 2.

Detail 1: Crop-specific comparisons: The crop-specific comparison extends to whether the crop followed a legume crop or a non-legume crop. This is done to ensure a fair comparison of practices. For example, if in the project-scenario a corn crop followed a soy crop, then the correct crop from the historical period to select for comparison should also be a corn crop followed by a soy crop. When the preceding crop of a baseline crop is not known then that crop could be considered included in all potential comparisons.

Detail 2: New crops in rotation: One of the defined additional management practices is the incorporation of a new cash crop into the rotation. In this case, the additional practice is evaluated for the entire crop rotation (i.e., is a new crop planted that wasn't planted at all in the historical period). New crops in rotation may include a new sequence of crops relative to a legume planting because management in a season following a legume crop may differ from management of the same crop in a season not following a legume. For example, a corn crop planted following a soy crop may have completely different nitrogen management practices as compared to a corn crop planted following a corn crop or other non-legume crop.

Table 2: Additional practice quantification approaches for the historical period

	Crop-specific Comparison	**Rotation Comparison**
When used	Index cultivation cycle cash crop is also grown during the historical management period	Index cultivation cycle cash crop is new to the historical rotation
Applicable additional management practices	• All tillage practices • New cover crop adoption • Longer duration of cover crops • Adding new species to existing cover crop • All nitrogen application practices	New cash crop in rotation
Comparison of quantitative values	Management practice is quantified as the simple **average** for each cultivation cycle	

Step 4 - Compare and evaluate the quantified practices across the index cultivation cycle and the historical period

Once these practices have been quantified in both the index cultivation cycle and the appropriate historical comparison period, the proposed additional management practice is evaluated against the collected historical information with the thresholds pre-defined for each additional management practice. Each threshold was determined with agronomic consultation and approval before use.

If the delta between the quantified values of the index cultivation cycle and the historical period was not within the bounds of the additional management practice threshold, then the evaluated management practice would not quality and therefore would not be deemed "additional".

Exclusion criteria:

There is one unique circumstance that must be checked for before fields move on to Step 5. As defined in Section 6.5.1 of the SEP, fields which employ no-till in the project scenario after continuously tilling in their historical period (to depths deeper than 20cm) will not be eligible to be credited for SOC gains. These fields can only be eligible if/when they subsequently adopt any tillage practice other than no-till. Based on this guidance, Indigo has included exclusionary checks with the following logic:

- Check for "continuous deep tillage" as defined as tillage more than 20 cm (based on the effective depth of disturbance, listed in Table 7 of `IndigoCarbon_US-1_2023_0030)` in more than one year of the historical period, regardless of the crop.
- If a field's historical period includes "continuous deep tillage", then the field may not be included in the Project unless (or until) they till in the project scenario.

Importantly with this construction, when the field does become eligible (with a season with a tillage event), the field's historical period will start in the years immediately before its first eligible season. Therefore, the baseline may not reflect the period of time when the grower employed continuous deep till.

Step 5 - Seek attestation from the grower if an additional practice is identified that was not attested by the grower

In cases where the software identified a qualifying practice that the grower did not indicate when entering his/her data for that period, Indigo will contact those growers to confirm that the change was in fact legitimate and intentional – regardless of whether the attested practice was confirmed or not. If the grower attests to this additional practice, Indigo's data requirement standards will apply to the new practice which may lead to additional data requirements by the grower.

Step 6 – If no additional management practice change qualifies in the current cultivation cycle, repeat process with the next potential cultivation cycle

The process described above will be duplicated until either 1) an additional management practice is identified and attested to by the grower, or 2) until all potential cultivation cycles have been assessed during the qualifying years.

For fields which meet the requirements of this test, their additional management practice(s) will not have to be reassessed until the crediting period for that field is being renewed.

3 Subsequent years of the additional management practice assessment

For fields which meet the requirements in this test, their additional management practice(s) will not have to be reassessed until the crediting period for that field is being renewed.

4 Additional management practice assessment spatial granularity

The assessment of additional management practices was conducted on a field basis (assuming 100% of a field's acres) or on a management zone basis (assuming a proportion of the field's acres). The adoption of an additional management practice on a proportion of a field was considered if the practice change on the management zone was significant enough to affect the management on the entire field. Below are a few examples to demonstrate this criterion:

- **Example:** if a field is transitioned from medium till on the entire field to no-till on 80% of the field and medium till on the remaining 20% of the field (to break up a compaction layer in a low-laying portion of the field), the

field was still considered to meet additionality because the overall effect of this management-zone-level change was to reduce the field from medium tillage to low tillage.

Furthermore, this approach ensures continuity of the period of time over which baselines were constructed for a field. If there were only parts of a field allowed to be additional (and therefore eligible for credit generation), then a scenario would be created where another part of the field could qualify for the Project in a subsequent reporting period which would break up the historical period for that field. Following the requirements of the SEP, Indigo sought to evaluate additionality on a field basis and developed the use of management zones to enable this result.

5 Summary of Changes

Section	Date	Summary of Changes
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